17 An electric current of 12 A is in a wire of length 0.35 m.

The average drift speed of the free electrons (charge carriers) in the wire is 5.0×10^{-4} m s^{-1}.

How many free electrons are in the wire?

A 1.3×10^{16} **B** 5.3×10^{22} **C** 1.5×10^{23} **D** 4.3×10^{23}

Ans: B $I = nAvq \Rightarrow \left(\frac{N}{V}\right)Avq = \frac{N}{L}vq \Rightarrow N = \frac{IL}{vq} = \frac{(12)(0.35)}{(5.0\times10^{-4})(1.60\times10^{-19})} = 5.3\times10^{22}$